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2025_02

Proposal Document:
INDOOR ENVIRONMENT MONITORING SYSTEM.

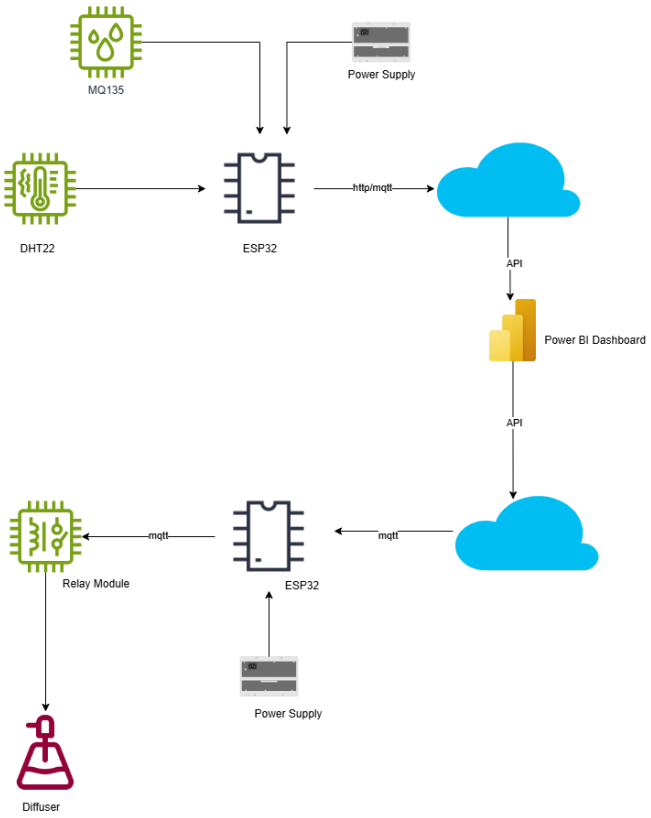
IT4021 : Internet Of Things and Big Data Analytics

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1. Project Description.

Indoor Environment Monitoring (IEM) system focuses on implementing an IoT-based environmental monitoring system that collects real-time data on the Ethane, Propane and Carbon Monoxide(CO) levels, Humidity and Temperature of the Indoor environment. The collected data is then stored and analyzed, after which the results and predictions of the analysis is displayed in a dashboard making the Results and predictions easily understandable. Alerts for Ethane, Propane and CO levels will be displayed in the dashboard. In addition, If the Humidity level in the environment is lower than the optimal level a diffuser will be triggered to on state, which could be manually switched Off through the dashboard or it will automatically switch of based on the humidity level. With all of these functionalities, the system will assist in air quality monitoring and environmental controls and assessment [1] [2] [3].

2. Architecture Diagram and Description.



The system is built on an IoT Framework consisting of multiple components where each component interacts with the external environment and each other to collect, process and visualize the environmental data. The Architecture of the system is as follows:

The DHT22 sensor [1] will collect data on the temperature and humidity levels of the environment, while the MQ135 sensor will collect data on the levels of Ethane, Propane and CO [2]. The sensors are connected and powered by a dedicated powers supply and also connected to an ESP32 Microcontroller which collects data from the sensors and transmits it to the cloud via MQTT protocol. The data stored in a cloud NoSQL database is integrated into a Power BI dashboard via API calls for real-time visualization and analysis.

Based on the results of the analysis, where incase of high levels of Ethane, Propane and CO levels the user is alerted through the dashboard and incase of low-levels of humidity an EPS32 microcontroller which controls a Relay module which in turn connects with a diffuser [3] which would then be switched on. The diffuser will automatically switch off once the humidity levels reach optimal levels. Also, for practicality and ease of use the dashboard will contain a power off button for the diffuser, which the user can use to manually switch off the diffuser based on the situation.

The architecture enables real-time monitoring, cloud-based data analysis and automated environmental control making a robust but easy to use Iot-Based environmental monitoring system.

3. Member Contributions

I.Herath H. M. T. N.

Uses MQ135 sensor to collect data related to CO ,Ethane and Propane levels in the environment and passes it to the dashboard performing the analysis to detect anomalies and generate alerts to users about gas leaks. The Member will also contribute with the diffuser configurations and assembly.

II.Jayawardena K. M. S. P.

Uses DHT22 sensor to collect data related to Humidity and temperature in indoor environments passes it to the dashboard where abnormally low levels of humidity and configure the diffuser to switch on based on the humidity and also have a function where the user can switch off the diffuser based on their preferences.

4. List Of Hardware.

I.Sensors.

- MQ135 Multi-Gas Sensor.
- DHT22 Temperature and Humidity sensor.

II.Actuators.

- 5V Relay Module.
- Diffuser.

III.Other Devices.

- 2 ESP32 Microcontrollers.
- Power supply : USB cables

5. Cost Breakdown.

Hardware Component	Number of Units	Cost (Rs.)
MQ135	1	370.00/-
DHT22	1	625.00/-
Relay Module	1	220.00/-
Diffuser	1	2000.00/-
ESP32 Microcontroller	2	4800.00/-
USB cables (Power supply)	3	1800.00/-
	9	9790.00/-

6. References

[1] S. Walling, J. Sengupta and S. Das Bit, "Low Cost Air Quality Monitoring System Using LoRa Communication Technology," in *IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, Goa, India, 2019.

[2] J. M. S. Waworundeng, "Design Prototype Detector of Temperature, Humidity, and Air Quality using Sensors, Microcontrollers, Solar Cells, and IoT," *Cogito smart journal,* vol. 9, no. 2, pp. 411-421, 2023.

[3] A. Dewangga, R. Muhammad, A. Sunardhi, W. Himawan and F. A. Zaky, "Development of an Internet of Things-based Environmental Information System for Realtime Monitoring of Air Conditions,," *Journal of Geography Science and Education,* vol. 6, no. 1, 2024.